How is the periodic table arranged?

The **periodic table** is arranged like a grid. The elements appear in order of increasing **atomic mass** and are placed in a specific location because of the way they look and act.

Period	Group 1	2										3	4	5	6	7	8 (or 0)
1	1 H																2 He
2	3 Li	4 Be										5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg										13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Uub	113 Uut	114 Uuq	115 Uup	116 Uuh	117 Uus	118 Uuo

57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

Rows

- The elements are placed in rows. All horizontal rows are called **periods.**
- All elements in the same period have the same number of **electron shells**.
- There are seven periods, but elements 57–71 and 89–103 have been placed underneath to make the table a more manageable shape.

Columns

- Elements are arranged in vertical columns called **groups** (1–8), with the **transition metals** between groups 2 and 3. For example, group 1 consists of H, Li, Na, K, Rb, Cs, Fr.
- The transition metals in the centre section behave a bit differently, so are not included.
- All elements except hydrogen in group 1 share similar properties. Hydrogen is a special element, so is often placed above group 1.
- The group number also represents the number of **electrons** in the **outer shell**. This number affects how **reactive** the element is. The most reactive elements have very few electrons in their outer shell, or only need a few more to fill it.

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ISBN 978 1 92571 062 5

Organising the periodic table

Read the Resource sheet 'How is the periodic table arranged?', then answer the questions.

☞ **1** What do we call the columns? _____

2 How many periods are there on the periodic table? _____

3 Which group does Cl (chlorine) belong to? _____

4 In which period is Cu (copper) found? _____

5 Circle the correct ending for each of the sentences below.

The horizontal rows are called	groups.	periods.
The groups run	from left to right.	from top to bottom
All elements in the same period	have the same number of electrons.	have the same number of electron shells.
All elements in the same group	have the same number of electrons in their outer shell.	have the same number of electron shells.

6 Decide whether each of these statements is true or false.

a The element C (carbon) is found in period 3. _____

b The element K (potassium) is found in group 1. _____

c Na (sodium), P (phosphorus) and Cl (chlorine) atoms all have 4 electron shells. _____

d Mg (magnesium), Ca (calcium) and Ba (barium) all have 1 electron in their outer shell. _____

7 Complete the table below.

Element	Number of electrons	Number of electron shells	Number of electrons in outer shell
Mg (magnesium)			
I (iodine)			
Al (aluminium)			

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